

HACK CORE 2026

MASTER RESEARCH, API, DATASET & HACKATHON STRATEGY GUIDE

CE Hub + Syngenta data ecosystem + supplied India algorithm + previous hackathon research

Purpose

A single working document to understand what data is available, what should be used, how the APIs fit the algorithm, what Syngenta-specific data can make the solution stronger, and what previous Syngenta/ANNAM.AI hackathons suggest about winning project strategy.

Core thesis: Do not build a generic weather dashboard. Build an explainable agricultural decision engine that uses the CE Hub ecosystem to transform historical + current + forecast signals into crop-specific biological recommendations.

Prepared for the Ideation Phase • Deadline: 16 August 2026, 5:00 PM

1. Executive Summary

The supplied CE Hub presentation describes a platform that brings together historical, real-time, nowcast and forecast weather data and turns it into decision-ready services for field, agronomy and digital-product teams. It explicitly identifies system APIs and actionable insights such as disease risk, weather alerts and spray windows. [C1]

The supplied algorithm document then defines the actual decision logic for India: Stress Buster and Yield Booster apply to Rice, Wheat and Cotton, while the document also defines yield/nutrient-related calculations. [A1]

The strongest technical direction is therefore a pipeline rather than a chatbot:

CE Hub / approved data → data normalization → crop-specific algorithm → risk scores → explainable recommendation.

For the MVP, prioritize only variables that directly drive the supplied formulas. Add satellite/NDVI, station-level validation and richer real-time intelligence only where access is confirmed and the added value is demonstrable.

2. What the CE Hub Presentation Tells Us

CE Hub is described as a single access point for weather intelligence. It combines historical, real-time and forecast weather data, then converts the information into decision-ready services. [C1]

Data horizon	What the supplied deck says	Potential role
Historical	Gridded datasets + station records; records dating back to 1940.	Historical baselines, risk analysis, validation.
Real-time	Current weather conditions and measurements where available.	Current-state monitoring and alerts.
Nowcast	Very short-term predictions for the next 0–6 hours.	Immediate operational decisions.
Forecast	Short and medium range; 1–7 days and 7–14 days.	Future risk and planning.

The deck says APIs are intended for programmatic access and product integration/automation, while portals are intended for exploration and self-service workflows. [C2]

3. Syngenta / CE Hub Historical Data — What Is Available?

The CE Hub presentation identifies two complementary historical routes.

Source	Details from supplied deck	Value to our project
Historical gridded data	Records back to 1940; 30+ datasets from providers such as ECMWF, DWD, NOAA and Meteoblue; simulation, reanalysis and satellite-derived datasets.	★★★★★ Core source for historical crop/weather analysis.
Weather station data	170,000+ stations worldwide; 30,000 Syngenta-owned stations and 140,000+ public stations; temperature, precipitation, humidity, wind, pressure and more.	★★★★★ Excellent for local validation if access is granted.

Important distinction: the presentation establishes that these resources exist in the CE Hub ecosystem. It does not by itself guarantee that every station, dataset or variable is exposed to HACK CORE teams. Exact access must be confirmed in Swagger or with the mentor.

4. Forecast Data — What We Should Use

The CE Hub deck describes operational forecast data with a 1–7 day short-range horizon and a 7–14 day medium-range horizon. Forecast parameters listed are temperature, precipitation, humidity, wind, solar radiation, cloud cover, pressure and derived evapotranspiration. [C3]

Variable	Direct use in supplied algorithm	Strategic use
TMAX	YES — daytime heat stress; GDD	Core forecast risk.
TMIN	YES — nighttime heat; frost; GDD	Core forecast risk.
Precipitation	YES — drought; yield; nutrient factors	Core forecast risk.
ET / evaporation	YES — drought index	Core drought signal.
Soil moisture	YES — drought + nutrient	Verify exact source/access.
Humidity	No	Potential disease model / future feature.
Wind	No	Potential spray-window feature.
Solar radiation	No	Potential crop-growth feature.
Cloud cover	No	Potential operational context.
Pressure	No	Low MVP priority.

5. Supplied India Algorithm — What It Requires

For India, the supplied algorithm document lists Stress Buster and Yield Booster for Rice, Wheat and Cotton. [A1]

Algorithm	Required inputs	Data class
Day heat stress	TMAX	Weather
Night heat stress	TMIN	Weather
Frost stress	TMIN	Weather
Drought risk	Rainfall P, ET/evaporation E, soil moisture SM, average temperature T	Weather + soil
Yield risk	GDD, rainfall, pH, available N	Weather + field/soil
Nitrogen stress	Projected yield, N applied, rainfall, soil moisture	Weather + farm
Phosphorus stress	Yield, P applied, pH factor, soil moisture factor, rainfall factor	Weather + farm/soil

This table is the key to deciding which API variables matter. Everything else should be treated as optional until the core pipeline works.

6. India Crop Thresholds

Crop	TMax optimum	TMax limit	TMin optimum	TMin limit	Frost
Rice	32°C	38°C	22°C	28°C	NA
Wheat	25°C	32°C	15°C	20°C	NA
Cotton	32°C	38°C	20°C	25°C	4°C to –3°C

The supplied document defines heat-stress scores on a 0–9 scale. For daytime heat, the score is 0 at/below the optimum, 9 at/above the limit, and linearly interpolated between. [A2]

7. Core Formulas to Implement

Daytime heat stress

$TMAX \leq TMaxOptimum \rightarrow 0$; $TMaxOptimum < TMAX < TMaxLimit \rightarrow 9 \times (TMAX - TMaxOptimum) / (TMaxLimit - TMaxOptimum)$; $TMAX \geq TMaxLimit \rightarrow 9$. [A2]

Nighttime heat stress

$T_{MIN} < T_{MinOptimum} \rightarrow 0$; $T_{MinOptimum} \leq T_{MIN} < T_{MinLimit} \rightarrow 9 \times (T_{MIN} - T_{MinOptimum}) / (T_{MinLimit} - T_{MinOptimum})$; $T_{MIN} \geq T_{MinLimit} \rightarrow 9$. [A2]

Frost stress

For applicable crops, the supplied document uses 4°C as the no-frost threshold and -3°C as the severe threshold. Rice and Wheat are marked NA, so do not invent thresholds. [A3]

Drought risk

$DI = (P - E) + SM / T$. P = cumulative rainfall, E = cumulative evaporation/ET, SM = soil moisture, T = average temperature. [A4]

Yield risk

$YR = w1(GDD - GDD_{opt})^2 + w2(P - P_{opt})^2 + w3(pH - pH_{opt})^2 + w4(N - N_{opt})^2$. Example weights in the supplied document are 0.3, 0.3, 0.2, 0.2. $GDD = [(T_{max} + T_{min}) / 2] - T_{base}$. [A5]

Nitrogen / phosphorus efficiency

$NUE = (Crop\ yield / Nitrogen\ applied) \times Rainfall\ factor \times Soil\ moisture\ factor$. $PUE = (Yield / P\ applied) \times Soil\ Factor$. The supplied PUE section defines SF using pH, soil-moisture and rainfall factors. [A6]

8. Data We Must NOT Assume Comes From the Weather API

Several required algorithm inputs are not simply weather variables. The exact availability of these fields through the supplied APIs must be verified.

Input	Required by	Safe assumption?	Plan
Soil pH	Yield + phosphorus	NO	Verify source; otherwise field input.
Available nitrogen	Yield	NO	Soil test / farm input.
N applied	Nitrogen stress	NO	Farm input.
P applied	Phosphorus stress	NO	Farm input.
Projected yield	N/P stress	NO	Farmer/model input.
Soil moisture	Drought + N/P	NO	Verify exact API variable or sensor input.
Organic matter	Mentioned in P discussion	NO	Verify if needed; not directly in displayed PUE equation.

9. Meteoblue Dataset API — How to Use It

Meteoblue's official Dataset API documentation describes a flexible JSON-based interface for querying weather/environmental datasets. It supports more than 100 variables from more than 50 data sources, but an individual dataset supports only a subset of all variables. [M1]

Recommended workflow:

- 1 Choose the field coordinates.
- 2 Choose the historical time interval.
- 3 Choose the appropriate dataset/domain.
- 4 Choose the required variable codes and levels.
- 5 Choose daily or hourly resolution.

- 6 Choose units.
- 7 Generate/test the query and export it.
- 8 Map the response into a normalized internal schema.

Generic request structure (template only):

POST https://my.meteoblue.com/dataset/query?apikey=YOUR_API_KEY

```
{ "units": { "temperature": "celsius", "velocity": "km/h", "length": "metric"}, "geometry": { "type": "MultiPoint",
"coordinates": [[LONGITUDE, LATITUDE, ELEVATION]] }, "format": "json", "timeIntervals": ["START/END"], "queries":
[{"domain": "DATASET_NAME", "timeResolution": "daily", "codes": [{"code": VARIABLE_CODE, "level":
"VARIABLE_LEVEL"}]} ] }
```

Do not guess dataset names or variable codes. Use the Dataset API configurator/documentation to select a dataset that actually contains the variables required by the algorithm. API credentials must remain in backend secrets/environment variables.

10. Syngenta CE Hub Forecast API — How to Use It

The supplied CE Hub deck explicitly positions APIs as the route for product integration and automation and points to the CE Hub Swagger/API documentation. [C2]

- 1 Open the supplied Swagger URL.
- 2 Find forecast endpoints.
- 3 Record authentication requirements.
- 4 Record required location and time parameters.
- 5 Record forecast horizon and temporal resolution.
- 6 Record exact response variable names and units.
- 7 Use Swagger's Try it out feature if permitted.
- 8 Save one request/response example for integration tests.
- 9 Implement the connector on the backend, never in public frontend code.

The Swagger UI itself was accessible during research, but the endpoint definitions were not machine-readable in the review environment. Therefore this guide intentionally does not invent endpoint names or schemas. The exact endpoint contract must be copied from the Swagger UI by the team.

11. What Data Should We Use First?

Priority	Data	Why
P0	TMAX + TMIN	Core heat/night/frost/GDD inputs.
P0	Rainfall	Drought + yield + nutrient factors.
P0	ET / evaporation	Drought formula.
P0	Soil moisture	Drought + nutrient factors; verify exact source.
P0	pH, N, N applied, P applied, projected yield	Required non-weather inputs.
P1	Syngenta station observations	Hyper-local validation if access is granted.
P1	Humidity / wind / solar / cloud	Useful future agronomic/spray intelligence.
P2	Other environmental datasets	Avoid unnecessary complexity in MVP.

Priority	Data	Why
P2 / WOW	NDVI / satellite imagery	Strong crop-condition layer if access is confirmed.

12. The 'Syngenta-Preferred' Data Angle: Station + Satellite

The strongest Syngenta-specific story is to prefer CE Hub data and then add other Syngenta ecosystem signals only when the challenge grants access.

Weather stations

The CE Hub deck reports 30,000 Syngenta-owned stations within its 170,000+ station ecosystem. This can be a powerful validation layer because it lets the project compare gridded/forecast values with local station observations where available. [C4]

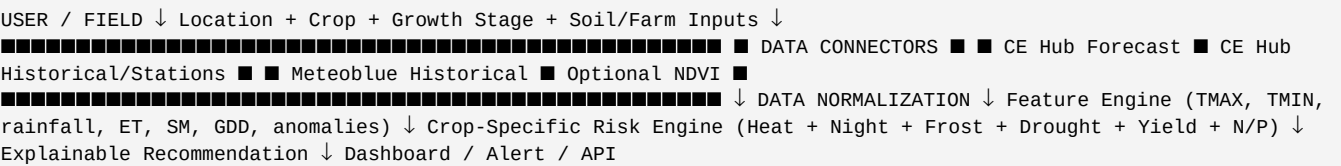
Satellite / NDVI

A previous Syngenta & CE Hub START Global 2024 hackathon guide explicitly exposed Cropwise Imagery with NDVI and visual satellite images for farm-drawn fields, including image download and geographic analysis workflows. This proves that such an integration has been used in a previous Syngenta hackathon, but it does not prove that HACK CORE 2026 teams have access. [S1]

Mentor question to ask: “Is Cropwise Imagery / NDVI or any Syngenta field-imagery layer available to HACK CORE 2026 teams?”

13. Recommended System Architecture

A strong architecture keeps external APIs separate from the decision engine.



The key design principle is explainability: every recommendation should be traceable to the input weather/soil signal and the algorithm rule that triggered it.

14. Example That Can Impress Judges

Scenario: Rice field in India.

- 1 Historical CE Hub/Meteoblue data establishes the local temperature and rainfall baseline.
- 2 Current observations show the field's present conditions.
- 3 CE Hub forecast predicts TMAX/TMIN and rainfall for the next 5–7 days.
- 4 The system calculates Rice daytime and nighttime stress using the supplied thresholds.
- 5 Drought risk is calculated when rainfall, ET, soil moisture and temperature are available.
- 6 Historical anomaly is computed to answer whether the forecast is unusual for this location/season.
- 7 The recommendation layer explains the dominant stress and whether the supplied biological intervention logic is triggered.

Example output:

HIGH HEAT-STRESS RISK
Rice daytime stress: 5.4/9

Night stress: 6.0/9
Forecast persistence: High
Historical anomaly: +3.1°C

Reason: forecast temperature remains above the crop's optimum threshold for multiple days.

Action: apply the challenge-approved Stress Buster recommendation logic.

The exact numerical example above is illustrative; the final system must calculate values from actual API responses.

15. Previous Syngenta Hackathon — START Global 2024

Syngenta's official case study says the company participated in START Hack for the first time in 2024. Twenty teams tackled the Syngenta case, working with rich historical and live data. The challenge centered on modelling the impact of farmer choices. [S2]

Winner: GIT HAPPENS

The team combined computer science, satellite imaging, AI and robotics to create HARVEST, a tool designed for governments to help smallholders. It provided parcel-level information, crops grown, estimated yields and advance disease warnings. [S2]

The strategic lesson is important: they did not build a generic farmer chatbot. They reframed the problem at a larger system level — effectively treating government as the layer that encompasses many farms — and focused on helping smallholders who can face high technology costs. [S2]

This previous winner also strengthens the case for asking whether satellite/NDVI and parcel-level data can be used in HACK CORE 2026.

16. Previous ANNAM.AI Agriculture Hackathon

ANNAM.AI's public site has a 2025 hackathon archive with project write-ups covering teams such as KRISHIVRA, AgroSaarthi, FASALGEO, CRISPRoots AgroTwin Nexus and others. [N1]

The archive demonstrates a broad range of agriculture-AI themes: AI farming assistants, offline AI, satellite/geospatial field analysis, precision agriculture and agricultural biotechnology concepts. [N1]

Because the currently accessible official archive does not expose a single consolidated winners table in the page content retrieved for this report, this document does not treat unofficial winner rankings as verified. The safe conclusion is that ANNAM.AI has already run a substantial agriculture-focused hackathon ecosystem and publishes team/project work publicly. [N1]

This is useful strategically: aim for a technically grounded, research-oriented, deployable agricultural system rather than a superficial UI demo.

17. Why Syngenta + ANNAM.AI Collaboration Matters

On 15 June 2026, Syngenta announced an MoU to become a strategic partner in the ANNAM.AI program. Syngenta said it would contribute agronomic and R&D; expertise to building accurate crop-health, pest-forecasting and heat-stress models. [S3]

Syngenta's announcement describes the broader vision as free, real-time and personalized agricultural intelligence at scale, with crop intelligence, real-time microclimate data and multilingual engagement. [S3]

That direction strongly overlaps with the HACK CORE resources: weather intelligence + crop-specific models + actionable field recommendations.

18. What We Should Build to Stand Out

Avoid positioning the product as simply:

“An AI chatbot for farmers.”

A stronger positioning is:

“An explainable, crop-specific biological decision intelligence layer built on CE Hub weather intelligence, historical baselines and future forecasts.”

Layer	Our differentiation
Data	Prefer CE Hub/Syngenta ecosystem data first.
Historical intelligence	Establish local/seasonal baseline rather than using forecast alone.
Forecast intelligence	Predict upcoming risk windows.
Agronomy	Use the supplied crop-specific formulas exactly where specified.
AI	Use ML only where it adds calibration, anomaly detection or uncertainty — not as a replacement for the supplied rules.
Explainability	Show why a recommendation happened.
Scalability	Design around locations/fields so it can serve many farms.
Impact	Measure avoided stress, improved timing, resource efficiency or yield-risk reduction where defensible.

19. Questions That Will Impress the Mentor

- 1 Which exact CE Hub forecast endpoints should HACK CORE teams use?
- 2 Which historical CE Hub dataset should be considered the preferred source for TMAX/TMIN/rainfall/ET?
- 3 Can we access Syngenta-owned weather station data for locations where stations exist?
- 4 Is soil moisture available through CE Hub, and what is the authoritative variable/endpoint?
- 5 Is Cropwise Imagery / NDVI available for this challenge?
- 6 Where should pH, nitrogen, phosphorus application and projected yield come from?
- 7 What is the correct Tbase for GDD for Rice, Wheat and Cotton?
- 8 The drought section labels both $DI = 1$ and $DI < 1$ as medium risk; what is the intended boundary?
- 9 The phosphorus Soil Factor uses three factors but divides by 4; is that intentional?
- 10 Should the supplied deterministic algorithm be reproduced exactly, or can historical Syngenta data be used to calibrate thresholds/weights?
- 11 Should individual risk scores be presented separately or combined into one final recommendation score?

20. Development Roadmap

Phase	Work	Result
1	Inspect Swagger + confirm access	Exact endpoint contracts.
2	Connect historical API	Verified historical dataset.
3	Connect forecast API	Verified forecast dataset.

Phase	Work	Result
4	Normalize data	Common internal schema.
5	Implement Stress Buster	Heat/night/frost/drought scores.
6	Implement Yield/Nutrient	Yield + N/P risk where inputs exist.
7	Add historical anomaly + validation	More credible recommendations.
8	Optional station/NDVI layer	Syngenta-specific differentiation if access granted.
9	Dashboard + API + demo	Judge-ready MVP.
10	Backtest + scenario tests	Evidence that logic works.

21. Important Caveats / Things Not to Claim

- Do not claim access to all Syngenta-owned weather stations unless the mentor/API confirms it.
- Do not claim Cropwise/NDVI is available for HACK CORE 2026 merely because it was used in a previous Syngenta hackathon.
- Do not invent CE Hub endpoint names, API fields or authentication schemes.
- Do not invent Tbase values for GDD.
- Do not silently correct ambiguous formulas in the supplied algorithm; flag them for mentor confirmation.
- Do not treat every CE Hub/Meteoblue variable as necessary. Start with direct algorithm inputs.
- Do not put API keys in the GitHub repository.

22. Sources & Evidence Map

[C1–C4] Challenge-provided source: CE Hub – Overview Presentation supplied by the user. It describes CE Hub's platform, access model, historical datasets, station data, forecast data and weather-based insights. [C1] pp. 2; [C2] p. 3; [C3] p. 6; [C4] p. 5. File evidence: [\[filecite\]turn0file0\[L12-L35\]](#); [\[filecite\]turn0file0\[L39-L59\]](#); [\[filecite\]turn0file0\[L86-L100\]](#); [\[filecite\]turn0file0\[L103-L119\]](#)

[A1–A6] Challenge-provided source: Algorithm Logic Document supplied by the user. India products, heat stress, night stress, frost, drought, yield risk and nutrient logic are defined across pp. 1–9. File evidence: [\[filecite\]turn0file1\[L17-L39\]](#); [\[filecite\]turn0file1\[L42-L68\]](#); [\[filecite\]turn0file1\[L78-L100\]](#); [\[filecite\]turn0file1\[L103-L125\]](#); [\[filecite\]turn0file1\[L134-L175\]](#); [\[filecite\]turn0file1\[L176-L212\]](#); [\[filecite\]turn0file1\[L213-L259\]](#)

[M1] Meteoblue official documentation: Dataset API overview and query documentation. <https://docs.meteoblue.com/en/weather-apis/dataset-api/overview>

[S1] Syngenta & CE Hub START Global 2024 guide: Cropwise Imagery, NDVI and visual satellite imagery, including image downloads and geographic analysis. <https://syngenta.github.io/syngenta-start-global-hackathon-2024/guide.html>

[S2] Syngenta Group official case study: START Hack 2024 Syngenta challenge, 20 teams, winner GIT HAPPENS and HARVEST. <https://www.syngentagroup.com/sustainability/stories/hacking-through-weeds>

[S3] Syngenta India official announcement: 15 June 2026 MoU with ANNAM.AI and focus on crop health, pest forecasting and heat-stress models. <https://www.syngenta.co.in/syngenta-signs-mou-be-strategic-partner-indias-first-open-data-agricultural-ecosystem-annamai>

[N1] ANNAM.AI official blog archive: public 2025 agriculture hackathon project write-ups. <https://annam.ai/blog/>

23. Final Recommendation

For the ideation submission, make the story very simple:

- 1) Use Syngenta/CE Hub first. Historical gridded data + forecast + station data where access is confirmed.
- 2) Use only the variables required by the supplied algorithm for the MVP. TMAX, TMIN, rainfall, ET and soil moisture are the core weather-side signals.
- 3) Collect/verify non-weather inputs. pH, nitrogen, fertilizer application and projected yield cannot be assumed to come from a weather API.

- 4) Add historical context. A recommendation should explain not only what the forecast says, but whether the coming conditions are unusual for the field/season.
- 5) Ask for Syngenta-specific layers. Weather stations and Cropwise/NDVI are the strongest questions to ask the mentor.
- 6) Keep the algorithm explainable. AI can enhance calibration/anomaly detection later, but the supplied agronomic rules should remain visible and testable.
- 7) Design for scale. The strongest precedent from Syngenta's 2024 hackathon winner was a system-level solution that could support many smallholders, not just a single-field demo. [S2]

This combination gives the project a clear identity: Syngenta data ecosystem + CE Hub weather intelligence + crop-specific agronomy + explainable AI + scalable field recommendations.